# Validation and AI use

**October 19, 2026 | Review copy**

## Decision, version and boundary

Proposed watch/defer for a committee with no SMR position. Analysis boundary: September 1, 2026 close, later than the pictured Edition A lab time. Research and draft construction occurred October 19. No September authorship is claimed. The working record matches the generated draft PDF when rebuilt.

Core environment: Python 3.10+ and standard library. PDF rebuilding uses optional packages in requirements-docs.txt. Input and source SHA-256 hashes are embedded in output/commercialization-output.json. The document source record provides file hashes; no LMS receipt is implied.

## Data and convention ledger

Historical observations, units, dates and filing locators are in docs/verified-starting-point-draft.md and data/research-inputs.json. These are reported June balances, June awards and July cover shares, not September balances. The model's assumed $40m lag-period cash use reduces cash and opening equity. The unrestricted liquidity sum adds each selected category once; restricted balances are outside it and are not subtracted a second time.

Current valuation uses consolidated economic interests with a gross award allowance. No book-NCI subtraction is made again. Gross options with no proceeds are an allowance, not a treasury-stock-method calculation. Equity funding affects liquidity; its fixed-price dilution diagnostic is separate from FCFF valuation.

Operating commitments and leases are assumed inside forecast costs. This requires a contract-by-contract mapping before a final value. Additional claims default to zero as a scenario condition, not a legal conclusion. The engine uses economic PMA expense timing, not GAAP revenue recognition or a tax opinion.

## Source check and assistant correction

The assistant opened the PMA exhibit, Sections 1(c), 1(d), 1(g), and found the annual escalation floor and redacted cap. Earlier flat-fee outputs were preserved and superseded. The revised fee scale is inferred from the 2025 cash installment and installment allocation. Future fees grow at the 5% floor; higher CPI is not forecast. Lead-year timing simplifies contractual slots, invoices and contingencies.

An unsupported low-growth perpetuity is disabled. Default operating exit value is six times positive normalized next-year EBITDA after full milestone fees. The multiple is an analyst assumption, not a peer benchmark. Zero/4x/6x/8x/10x alternatives are computed. Negative EBITDA receives zero operating sale value, not a liquidation appraisal. Quantitative peer validation remains unresolved.

## Validation register

| Claim | Check | Actual disposition | Evidence |
|---|---|---|---|
| Cash is not counted twice | No-operations known answer | PASS | tests/test_commercialization.py |
| Cash follows financial flows | Independent cumulative cash budget | PASS | same test file |
| Statements reconcile | Assets equal claims; equity rolls from income/funding | PASS | same test file |
| Sunk tranche is not paid twice | Single-cohort hand formula with escalation | PASS | same test file |

| Claim | Check | Actual disposition | Evidence |
|---|---|---|---|
| Contract escalation matters | Compare escalated vs flat fees | PASS | same test file |
| Funding is not double charged | Financing-price change affects diagnostic shares only | PASS | same test file |
| Margin stress worsens economics | Unfloored equity falls; funding does not fall | PASS | same test file |
| Discount dates are explicit | Recompute actual-day discount factors | PASS | same test file |
| Invalid assumptions fail | Reject bad rates, deliveries and nonfinite inputs | PASS | same test file |
| Unsupported terminal path fails | Reject growth below fee escalation if enabled | PASS | same test file |
| UI displays executed values | Compare CLI summary with frozen output | PASS | tests/test_application.py |
| UI detects stale evidence | Alter copied input and require error | PASS | same test file |
| Invalid year is actionable | Require nonzero exit and valid range | PASS | same test file |
| Old downside invariants | Cash budget, bridge, funding, claims, broken balance | PASS: five checks | tests/test_smr_proforma.py |

Run python3 verify_project.py. Current suite: 27 tests passed. Tests demonstrate these specific invariants, not the accuracy of commercial forecasts.

## Failure retained and diagnosed

After fee escalation was corrected, one margin test initially failed because it required strictly lower floored share value even when both outcomes were zero. Diagnosis: limited-liability flooring masks a worse underlying equity result. The corrected test requires lower unfloored equity, nonincreasing floored value, and no reduction in funding. No finance input was tuned to force a passing test.

## Changed-input results - AI-generated, not the locked exercise

The three five-percentage-point module-margin stresses and the full price/discount grid are preserved in output/commercialization-output.json. Stronger execution moves from $6.77 to $3.34/share including the assumed operating sale. The watch/defer proposal remains because contract pricing and the assumed exit are unverified, not because the automated test establishes a final price target.

## Reproducibility and access

evidence/clean-run.md records an actual assistant rehearsal in a fresh temporary directory with site packages disabled. Its command outputs and JSON comparisons are preserved. This is not a claimed outside-person cold run. A real reviewer must record their own first failure/resolution if the assignment requires it. Repository publication state is recorded in evidence/repository-status.json. Instructional-team and video access remain separate checks; they are not inferred from an assistant's successful git connection.

## Existing NuScale Lab 11 changed-input evidence

The exercise is not classified as absent. Existing records include a supplied prediction, input/model snapshot, changed cost-growth input, before/after output, and reconciliation. They are preserved in evidence/lab11-smr/ and mapped in docs/lab11-evidence-map.md. Cost growth changes from 0% to 2%; 2030 FCFE moves from -174.639m to -196.199m. The recorded explanation attributes the 21.560m decline to 20.028m extra costs and 1.532m lost investment income. Watch/defer remains.

The saved narrative does not independently establish the prediction-lock timestamp or AI boundary during the original run. This is a documentary check on existing work, not an automatic requirement to repeat it. Joby comparison evidence is not substituted for NuScale ownership. No timestamp or attestation is invented.

## Named AI-use register

Tool: Codex; exact model version was not exposed in the task surface. Interaction date: October 19, 2026. Material uses: workspace audit, source navigation, code, scenario assumptions, tests, generated outputs, memo/research/validation drafts, PDF layout, application interface and rehearsal scripts.

Assistant disposition: modified the flat-fee proposal after checking the contract; withdrew unsupported perpetual terminal credit; qualified every module-price and delivery assumption. Primary-source checks and deterministic arithmetic are assistant work. Student accept/modify/reject decisions and independent checks are still pending and must be recorded truthfully.

Inherited Week 5 work is preserved. Its original AI history must be supplied by the student. This record does not infer authorship from a test pass, file date, lab grade or code style. The user's account of grammar assistance is retained as their account, not independently verified history.

## Limitations, monitoring and kill rules

See docs/monitoring-policy.md. The full event/share audit, net contract economics, invoice timing, investment liquidity, authorized share capacity, contingent/tax claims and residual valuation remain open. Assumed equity issuance is not committed capacity, and end-year cash checks cannot establish intra-year solvency. No default, legal liability or certain loss is inferred from a model flag.

## Source references

PMA: https://www.sec.gov/Archives/edgar/data/1822966/000182296625000144/a101-partnershipmileston.htm June 10-Q: https://www.sec.gov/Archives/edgar/data/1822966/000182296626000085/smr-20260630.htm Source ledger and boundary review: docs/verified-starting-point-draft.md; docs/source-review.md.

## Live application and method verification

Seven additional tests check known-answer CAPM/capital weights, TTM and peer bridges, live recomputation and reset, invalid API inputs, fee/growth conflicts, terminal percentages, and a negative-cash-flow monotonicity counterexample. Total: 27 passing tests. The original 15% outputs remain reproducible; the proposed CAPM preset is separate. Browser verification is recorded in evidence/workbench-browser-checks.json when completed.

Codex also implemented the workbench and method comparison. Personal acceptance of beta, bounds and forecast choices remains outstanding; see assumption-register.md.